zeroriscy_register_file

OVERVIEW

The Register File is a compact and high-speed memory block within the processor, responsible for storing intermediate data during instruction execution. It serves as a temporary storage unit for operands, computation results, and control-related data required by the execution stages.

This module provides:

- Two simultaneous read ports
- One write port
- 32 general-purpose registers (x0 to x31)

To improve power efficiency, the design incorporates clock gating techniques, ensuring that only the required register receives the clock signal during write operations. Additionally, register x0 is permanently hardwired to zero, maintaining compliance with the RISC-V ISA specification.

1. READ OPERATIONS

The module supports two independent read operations simultaneously through:

- rdata_a_o
- rdata_b_o

This enables the processor to fetch two source operands in a single cycle, which is essential for efficient instruction execution.

2. WRITE OPERATION

A single write port is provided for updating register values. The write operation is controlled by:

- we_a_i -> Write Enable signal

When enabled, data is written into the selected destination register.

3. ZERO REGISTER HANDLING

Register mem[0], corresponding to register x0, is permanently fixed to zero. This means:

- It always outputs 0
- It cannot be overwritten

This behavior strictly follows the RISC-V architecture specification.

4. CLOCK GATING

To reduce unnecessary power consumption, clock gating is implemented so that the clock is enabled only during:

- Valid write operations
- Access to the targeted register

This significantly minimizes switching activity and improves overall power efficiency.

KEY FEATURES

- Dual-read and single-write architecture
- 32 registers with 5-bit addressing
- Register x0 permanently fixed to zero
- Clock gating for low-power operation
- One-hot write enable decoding
- Parameterized data width (DATA_WIDTH = 32)

INPUT AND OUTPUT SIGNALS

Inputs / Basic Control Signals:

- clk -> System clock
- rst_n -> Active-low reset
- test_en_i -> Test mode enable (bypass clock gating during testing)

Read Ports:

- raddr_a_i [4:0] -> Read address for Port A
- rdata_a_o [31:0] -> Read data output for Port A
- raddr_b_i [4:0] -> Read address for Port B
- rdata_b_o [31:0] -> Read data output for Port B

Write Port:

- waddr_a_i [4:0] -> Write address
- wdata_a_i [31:0] -> Write data
- we_a_i -> Write enable signal

OPERATION FLOW: READ OPERATION

The read operation is purely combinational, which means data is accessed directly without waiting for a clock edge.

Read Logic:

rdata_a_o = mem[raddr_a]

rdata_b_o = mem[raddr_b]

Advantages: No clock dependency, immediate data access, and faster processor performance.

OPERATION FLOW: WRITE OPERATION

A write operation occurs only when we_a_i = 1.

Write Sequence:

1. The write address is decoded into a one-hot signal.
2. Clock gating activates only the selected register.
3. Data is written on the gated clock edge.

This ensures accurate and power-efficient register updates.

ONE-HOT ADDRESS DECODING

The write address is converted into a one-hot encoded signal:

waddr_onehot[i] = (waddr == i)

Purpose: This guarantees that only one register is selected for writing and no accidental multiple writes occur. This improves reliability and design precision.

CLOCK GATING MECHANISM

The design uses two levels of clock gating for optimal power reduction:

1. Global Write Enable Clock: Enabled only when we_a_i = 1, preventing unnecessary clock activity during idle cycles.

2. Per-Register Clock Gating: Each register has its own gated clock. Only the selected register receives the active clock signal during a write operation.

Result: Reduced switching activity, lower dynamic power consumption, and efficient low-power processor design.

WRITE DATA BUFFERING

The signal wdata_a_q temporarily stores the write data before it is written into the register.

Purpose: This ensures stable data during gated clock operation and reliable write execution.

INTERNAL ARCHITECTURE

Memory Array:

- The register file consists of mem[0] to mem[31] (Total: 32 registers).
- Address Width: ADDR_WIDTH = 5. Since NUM_WORDS = 2^5 = 32, a 5-bit address is sufficient to access all registers.

Clock Gating Blocks:

- The design uses the cluster_clock_gating module for global write enable control and individual register-level clock control. This forms the foundation of the low-power architecture.

SUMMARY

The zeroriscy_register_file is a high-performance and power-optimized storage unit designed for efficient processor operation.

It supports simultaneous dual read operations, performs controlled single write operations, uses gated clocks to minimize power consumption, ensures correctness using one-hot write selection, and maintains strict RISC-V compliance through fixed register x0. This makes it a critical and highly optimized component of the Zero-Riscy processor core.